IGCSE 0607 Practice Questions — E1.15 — Combined Questions & Markscheme

Money

Total Questions: 9 — **Total Marks:** 53 Marks

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  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(7 Qs)*

1 John has \$400. He buys a new camera that costs 255 Euros (). The exchange rate is \$1 = 0.85.
Work out how much money John has left in dollars.

\$ ..... **[3]**

2 The exchange rate between pounds (£) and dollars ($) is £1 = $1.50.
Anna changes £300 into dollars.
She spends $210 on a tablet.
She changes the remaining dollars back into pounds at the same exchange rate.

Work out how many pounds she receives.

£ ..... **[3]**

3 A car dealer buys a car in Japan for 1 250 000 JPY.
The exchange rate is £1 = 145 JPY.

(a) Calculate the cost of the car in pounds (£).

£ ..... **[2]**

(b) The dealer pays shipping costs of £600.
He then sells the car in the UK for £12 500.

Calculate his percentage profit on the total cost of buying and shipping the car.

..... % **[3]**

4 Amira travels from the UK to Japan.
She changes 850 into Japanese Yen (JPY).
The bank charges a 2% fee on the amount exchanged.
The exchange rate is 1 = 180 JPY.

(a) Calculate the amount of JPY she receives.

..... JPY **[3]**

(b) While in Japan, she spends 115000 JPY.
She changes her remaining JPY back to pounds ().
The new exchange rate is 1 = 184 JPY.
The bank charges a fixed fee of 500 JPY for this transaction.
Calculate the amount of pounds she receives.

..... **[4]**

5 A trader buys 1000 items for y each.
He pays a shipping fee of 250.
He sells all 1000 items in the USA for \$4.20 each.

The exchange rate is 1 = \$1.20.
The trader makes a profit of 40% on his total cost in Euros.

Calculate the value of y.

$y =$ ..... **[5]**

6 In Year 1, the exchange rate between dollars ($) and euros (EUR) was $\$1 = x$ EUR.
In Year 2, the exchange rate changed to $\$1 = (x - 0.05)$ EUR.

In Year 1, a laptop cost 400 EUR. In Year 2, the same model laptop cost 450 EUR.

The cost of the laptop in dollars increased by $100 from Year 1 to Year 2.

(a) Write down an expression in terms of x for the cost of the laptop in dollars in Year 1.

$\$ \ldots\ldots\ldots\ldots\ldots\ldots$ **[1]**

(b) Form an equation in x and show that it simplifies to $20x^2 - 11x - 4 = 0$.

[4]

(c) Solve the equation to find the value of x.

$x = \ldots\ldots\ldots\ldots\ldots\ldots$ **[3]**

7 Amira goes on holiday to Japan.
She changes \$850 into Japanese Yen (JPY). The exchange rate is \$1 = 148 JPY.

(a) Calculate how many JPY she receives.

..... JPY **[2]**

(b) In Japan, she spends 45 000 JPY on a camera and 31 500 JPY on food.
At the end of the holiday, she changes her remaining JPY back to dollars (\$). The new exchange rate is \$1 = 152 JPY.
Calculate how many dollars she receives.
Give your answer correct to the nearest cent.

\$ ..... **[3]**

8 Carlos buys a car in the USA for \$17 400. He pays a shipping fee of \$1 200.

(a) The exchange rate is £1 = \$1.24. Calculate the total cost of the car and shipping in pounds (£).

£ ..... **[2]**

(b) Carlos sells the car in the UK for £17 100.
Calculate his percentage profit on his total cost.

..... % **[3]**

(c) Carlos changes his profit of £2 100 into euros.
The exchange rate is £1 = 1.16 euros.
The bank charges a 2% commission fee for the exchange.
Calculate the amount he receives in euros.

..... euros **[3]**

9 A shop sells laptops.
In 2022, the price of a laptop was \$840. In 2023, the price increased by 15%. In 2024, the price decreased by r% from the 2023 price. The price of the laptop in 2024 was \$821.10.

(a) Calculate the value of r.

$r =$ ..... **[3]**

(b) The shop buys 450 of these laptops from a supplier in China.
The total cost is 2 430 000 Chinese Yuan (CNY).
The exchange rate is \$1 = 7.2 CNY.

Calculate the profit the shop makes on each laptop when sold at the 2022 price of \$840.

\$ ..... **[3]**

(c) A customer in the UK buys one of these laptops in 2022 for \$840. They pay with a credit card that charges a 2.5% foreign transaction fee. The exchange rate is £1 = \$1.32.
Calculate the total amount the customer pays in pounds (£).
Give your answer correct to the nearest penny.

£ ..... **[3]**

1 $255 \div 0.85$ **[M1]**

$\$300$ **[A1]**

$\$100$ **[A1]**

[Back to Question] *[Total Marks: 3]*

2 $300 \times 1.50 = 450$ **[M1]**

$450 - 210 = 240$ **[M1]**

$240 \div 1.50 = 160$ **[A1]**

[Back to Question] *[Total Marks: 3]*

3 (a) $1\,250\,000 \div 145$ **[M1]**

8620.69 **[A1]**

(b) Total cost $= 8620.69 + 600 = 9220.69$ **[M1]**

Profit $= 12\,500 - 9220.69 = 3279.31$ (or $\dfrac{12500 - 9220.69}{9220.69}$) **[M1]**

35.6 (accept $35.56\ldots$) **[A1]**

[Back to Question] *[Total Marks: 5]*

4 (a) 850×0.98 or $850 - (850 \times 0.02)$ **[M1]**

833 **[A1]**

149940 **[A1]**

(b) $149940 - 115000$ $(= 34940)$ **[M1]**

their $34940 - 500$ $(= 34440)$ **[M1]**

their $34440 \div 184$ **[M1]**

187.17 (accept 187.17...) **[A1]**

[Back to Question] *[Total Marks: 7]*

5 Total revenue in USD $= 1000 \times 4.20 = 4200$ **[B1]**

Convert revenue to EUR: $4200 \div 1.20$ $(= 3500)$ **[M1]**

Equate to cost with profit multiplier: Total Cost $\times 1.40 = 3500$ or $3500 \div 1.40$ $(= 2500)$ **[M1]**

Form equation for cost: $1000y + 250 = 2500$ **[M1]**

$y = 2.25$ **[A1]**

[Back to Question] *[Total Marks: 5]*

6 (a) $\dfrac{400}{x}$ **[B1]**

(b) $\dfrac{450}{x-0.05} - \dfrac{400}{x} = 100$ **[M1]**

$450x - 400(x-0.05) = 100x(x-0.05)$ **[M1]**

$450x - 400x + 20 = 100x^2 - 5x$ **[M1]**

$50x + 20 = 100x^2 - 5x \implies 100x^2 - 55x - 20 = 0$ **[A1]**

Dividing by 5 yields $20x^2 - 11x - 4 = 0$ (must show working to reach final equation) **[A1]**

(c) $(5x-4)(4x+1) = 0$ or use of quadratic formula $\dfrac{11 \pm \sqrt{(-11)^2 - 4(20)(-4)}}{2(20)}$ **[M2]**

If M2 not scored, allow [M1] for $(5x \pm a)(4x \pm b) = 0$ where $ab = 4$ or $a(4) \pm b(5) = -11$, or correct substitution into formula but with one error.

0.8 (with -0.25 rejected or ignored) **[A1]**

[Back to Question] *[Total Marks: 8]*

7 **(a)** 850×148 **[M1]**

125800 **[A1]**

(b) $125800 - (45000 + 31500)$ or 49300 **[M1]**

$\dfrac{49300}{152}$ or 324.342... **[M1]**

324.34 **[A1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** $\dfrac{17400 + 1200}{1.24}$ or $\dfrac{18600}{1.24}$ **[M1]**

15000 **[A1]**

(b) $17100 - 15000$ or 2100 **[M1]**

$\dfrac{2100}{15000} \times 100$ **[M1]**

14 **[A1]**

(c) $2100 \times (1 - 0.02)$ or 2100×0.98 or 2058 **[M1]**

2058×1.16 **[M1]**

2387.28 **[A1]**

[Back to Question] *[Total Marks: 8]*

9 **(a)** 840×1.15 or 966 **[M1]**

$\dfrac{821.10}{966}$ or 0.85 **[M1]**

15 **[A1]**

(b) $\dfrac{2430000}{450}$ or 5400 **[M1]**

$\frac{5400}{7.2}$ or 750 **[M1]**

90 **[A1]**

(c) 840×1.025 or 861 **[M1]**

$\frac{861}{1.32}$ or 652.272... **[M1]**

652.27 **[A1]**

[Back to Question] *[Total Marks: 9]*